Weekly Report

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Overview

This week, I worked on building a complete OOK (On-Off Keying) free-space optical communication system and studied polarization controllers. The goal was to gain hands-on experience in laser modulation, circuit design, and signal reception.

OOK System Implementation

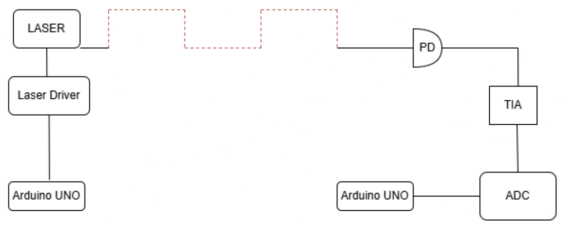


Figure 1: OOK Setup

Transmitter Side

An Arduino UNO was used to transmit a binary sequence (e.g., `"101011"`) using PWM. A laser driver circuit ensured safe operation of the laser diode by maintaining a current of 39.8 mA using a 100 Ω current sense resistor and a voltage divider for reference voltage generation.

Laser Driver Design

To regulate current through the laser diode safely and precisely, a laser driver circuit was implemented using an op-amp (LF347) and an NPN BJT in a feedback loop. A voltage divider generates a stable reference voltage of 2.0 V. This voltage appears across a 100 Ω resistor.
In stimulation i got 39.8 mA from the laser driver. the operating region of laser was 30 mA-50 mA.
The op-amp continuously adjusts the BJT base current to maintain this setpoint, providing closed-loop control. This ensures consistent brightness and protects the diode from current surges.

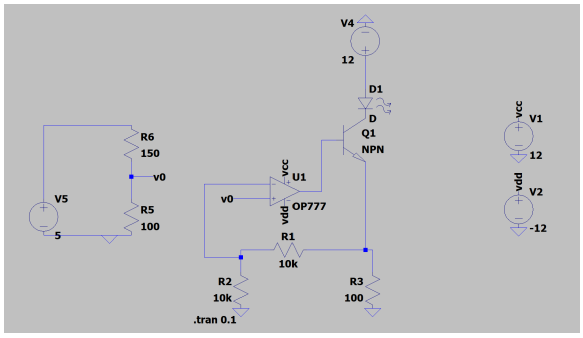


Figure 2: Laser driver stimulation on LT Spice

Receiver Side

A silicon photodiode detected the laser pulses, and a two-stage amplifier converted the photocurrent into a positive voltage output. The first stage was a TIA using an OP777 op-amp with 100 Ω feedback resistance and 5 pF compensation. The second stage inverted the signal back to positive voltage for ADC compatibility.
While trying to stimulate in LT Spice, i
tried stimulating with OP07 op-amp which really troubled me. I kept checking if everything else was correct or not and this ate up a lot of time.

Transimpedance Amplifier Design

The TIA converts the photodiode's current output into a voltage signal. An OP777 op-amp was used due to its low input bias current and high precision. A 100 Ω feedback resistor defines the transimpedance gain:

$$V_{\text{out}} = -I_{\text{photo}} \cdot R_f$$

A small capacitor (5 pF) in parallel with the resistor prevents high-frequency oscillations. As the TIA inverts the signal, a second identical op-amp stage inverts the signal again to yield a positive output. This two-stage setup provides stability, linearity, and compatibility with Arduino ADC input.

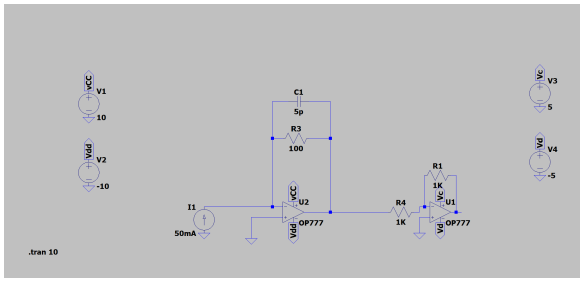


Figure 3: TIA Circuit stimulation with inversion in LT Spice.

PWM and Arduino

Arduino PWM was used to modulate the laser. Pins 3, 9, 10, 11 operate at 490 Hz, while pins 5 and 6 run at 980 Hz. Duty cycles were controlled using `analogWrite()` with values between 0–250.

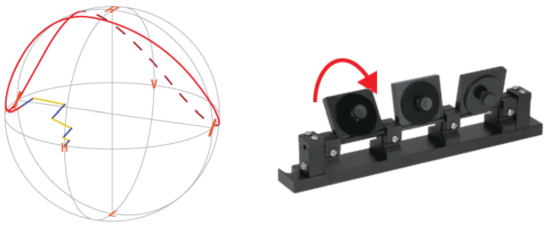
Understanding Polarization Controllers

During my presentation, sir, you asked me why we have three paddles and not two or more. You also told me to understand how they are made/designed.
I discovered that there are two-paddle and three-paddle polarization controllers.

Three-Paddle Polarization Controller:
This setup uses a quarter-wave plate (QWP), a half-wave plate (HWP), and another QWP in sequence. The first QWP converts the input to linear polarization, the HWP rotates it, and the final QWP sets the desired output state. Independent paddle control allows full manipulation of the polarization across a broad wavelength range (300 nm to 2100 nm).

Two-Paddle Polarization Controller:
This configuration uses two QWPs to adjust polarization. While it also provides full control, the tuning is more coupled and less intuitive, making it harder to achieve specific states. However, it is more compact and still supports broadband operation over 300 nm to 2100 nm.

We can understand it like -Stress in the fiber, alters its polarization, Stress on single mode fibers changes the polarization of light by bending and twisting optical fiber.



Each paddle has fiber loops that create different amounts of polarization change.
A common setup uses 2 loops on the first paddle, 3 on the second, and 2 on the third (2-3-2). This roughly mimics quarter-wave and half-wave plates.

Paddle Behavior on the Poincaré Sphere:

- **2-loop paddles:** Introduce polarization changes primarily in the *vertical direction*, i.e., along the axis connecting the circular polarization poles.
- **3-loop paddle:** Induces polarization displacement mainly in the *horizontal direction*, parallel to the linear polarization equator.